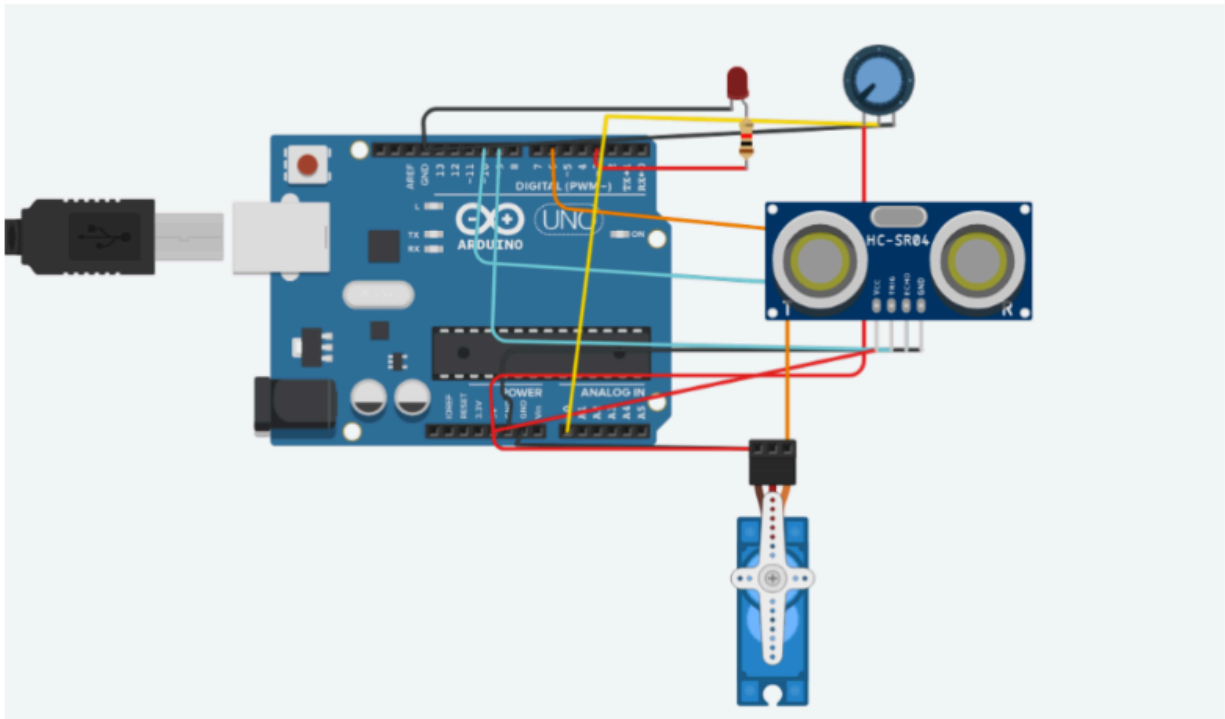




```
#include <Servo.h>
```

```
class GateSystem
```

```
{
```

```
private:
```

```
    Servo gate;
```

```
    int trigPin = 9;
```

```
    int echoPin = 10;
```

```
    int ledPin = 3;
```

```
    int potPin = A0;
```

```
    long duration;
```

```
    int distance;
```

```
    int threshold;
```

```
public:
```

```
    void setupSystem()
```

```
    {
```

```
        pinMode(trigPin, OUTPUT);
```

```
        pinMode(echoPin, INPUT);
```

```
        pinMode(ledPin, OUTPUT);
```

```
        gate.attach(6);
```

```

    gate.write(0);
    digitalWrite(ledPin, LOW);

    Serial.begin(9600);
}

void readDistance()
{
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);

    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);

    digitalWrite(trigPin, LOW);

    duration = pulseIn(echoPin, HIGH);
    distance = duration * 0.034 / 2;
}

void readThreshold()
{
    threshold = map(analogRead(potPin), 0, 1023, 5, 50);
}

void displayData()
{
    Serial.print("Distance: ");
    Serial.print(distance);
    Serial.print(" cm   Threshold: ");
    Serial.println(threshold);
}

void controlGate()
{
    if (distance <= threshold)
    {
        gate.write(90);
        digitalWrite(ledPin, HIGH);
    }
    else
    {
        gate.write(0);
        digitalWrite(ledPin, LOW);
    }
}

```

```
    }  
  }  
  
  void runSystem()  
  {  
    readDistance();  
    readThreshold();  
    displayData();  
    controlGate();  
  
    delay(100);  
  }  
};  
  
GateSystem myGate;  
  
void setup()  
{  
  myGate.setupSystem();  
}  
  
void loop()  
{  
  myGate.runSystem();  
}
```